C1.1 Enzymes and metabolism

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-c1-1.html

Written for this site from the class slides. The chapter is on pages 380–395 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Why enzymes matter

C1.1.1 Enzymes as catalysts

pp. 380–395

A **catalyst** speeds up a reaction without being used up. Without enzymes, many reactions would be too slow to sustain life. Heat would speed them up, but it would also damage proteins.

C1.1.2 The role of enzymes in metabolism

pp. 380–395

Metabolism is the complex network of interdependent reactions in an organism. Enzymes are specific, so organisms need many different ones. A cell can control metabolism through its enzymes.

C1.1.3 Anabolic and catabolic reactions

pp. 380–395

Anabolic: builds	Catabolic: breaks down
Protein synthesis, by condensation	Digestion, by hydrolysis
Glycogen formation	Oxidation of substrates in respiration
Photosynthesis	Decomposition

How enzymes work

C1.1.4 Globular proteins with an active site

pp. 380–395

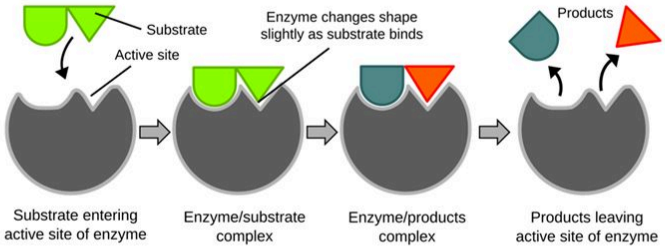
The **active site** contains only a few amino acids. Interactions between amino acids throughout the protein hold it in shape, and give it the properties needed for catalysis.

C1.1.5 Induced-fit binding

pp. 380–395

Induced-fit binding

C1.1.5



Induced fit. TimVickers, vectorized by Fvasconcellos, public domain.

As the substrate binds, **both enzyme and substrate change shape** to fit closely: **induced fit**. This strains bonds in the substrate so the reaction happens more easily.

Compare
The older lock-and-key model pictured a rigid active site. Induced fit explains the flexibility better.

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Both the substrate and the enzyme change shape as binding occurs. This places stress on bonds in the substrate.

C1.1.6 Molecular motion and collisions

pp. 380–395

Movement is needed for a substrate and active site to meet. In solution, random motion causes collisions. Large substrates can be immobilized, and enzymes can be fixed in membranes. In these cases, only the other molecule moves.

C1.1.7 Active site, specificity and denaturation

pp. 380–395

The active site has a shape and chemical properties that are complementary to the substrate, and this gives **specificity**. In **denaturation**, heat or extreme pH changes the active site, so the substrate can no longer bind.

Rates of reaction

C1.1.8 How temperature, pH and substrate affect the rate

pp. 380–395

Factor	Below the optimum, or low	Above the optimum, or high
Temperature	Rate rises, with more collisions	Rate falls sharply, due to denaturation
pH	Rate falls on either side of the optimum, as bonds holding the active site are disrupted	
Substrate	Rate rises, with more collisions	Rate levels off, as active sites are saturated

C1.1.9 Measuring reaction rates

pp. 380–395

Rate is the change in substrate or product per unit of time. A curve is steepest at the start, so use the **initial rate**. For example, 6 cm^3 of oxygen in 20 seconds is 0.3 cm^3 per second.

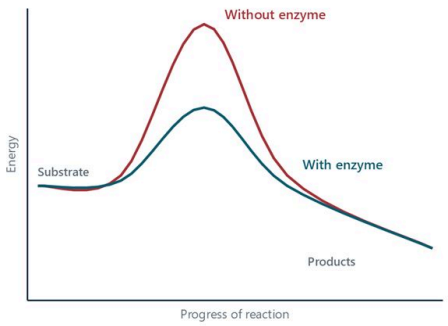
C1.1.10 Enzymes lower activation energy

pp. 380–395

Energy is needed to break bonds in the substrate, and energy is released as bonds form in the products. An enzyme lowers the **activation energy**, without changing the total energy released.

Enzymes lower activation energy

C1.1.10



Activation energy is needed to start a reaction. An enzyme **lowers** it, so many more collisions succeed at body temperature.

The energy of substrate and products is unchanged.

Common mistake

Enzymes do not add energy; they lower the barrier.

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